

Christian Cabrera-Jojoa

Assistant Research Professor

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I am an assistant research professor at the Department of Computer Science and Technology of the University of Cambridge and an Academic Associate at Pembroke College. I lead the [Interfaces research programme](#) at ML@CL on interpretable, self-sustaining multi-agent AI systems. I received a PhD in Computer Science from Trinity College Dublin in 2020. My prior research addressed resilient service-oriented systems for dynamic urban environments. Current projects include the [aICU programme](#) at Karolinska Institutet.

Education

- June 2020 **Ph.D. in Computer Science**, Trinity College Dublin, Dublin, Ireland
Dissertation: [uDiscovery: An Urban-Centric Model for Service Discovery in Smart Cities..](#) Supervisor: Prof. Siobhán Clarke.
- September 2014 **M.Sc. in Systems and Computer Engineering**, Los Andes University, Bogotá, Colombia
Dissertation: *MOWL: A Domain Specific Language for Handling Modular Ontologies.*
- September 2011 **B.Sc. in Systems Engineering**, Universidad de Nariño, Pasto, Colombia
- In progress **Postgraduate Certificate in Higher Education**, University of Cambridge, United Kingdom

Experience

- November 2025 **Assistant Research Professor**, Department of Computer Science and Technology, University of Cambridge, United Kingdom
- Current Lead the [Interfaces research programme](#) at ML@CL on interpretable, self-sustaining multi-agent AI systems, extending [the AutoAI project](#). Healthcare validation includes the [aICU programme](#) at Karolinska Institutet.
- November 2024 **Academic Associate**, Pembroke College, University of Cambridge, United Kingdom
- Current Undergraduate interviews and supervision at the college.
- October 2023 **Supervisor and demonstrator**, Department of Computer Science and Technology, University of Cambridge, United Kingdom
- Current *Artificial Intelligence, Software Engineering and Security, Concurrent and Distributed Systems, Databases, and C/C++.*
- March 2021 **Research Associate and Senior Research Associate**, Department of Computer Science and Technology, University of Cambridge, United Kingdom
- October 2025 Research on self-sustaining software systems (S4) and AI deployment as part of [the AutoAI project](#).
- October 2021 **Course director and lecturer**, Department of Computer Science and Technology, University of Cambridge, United Kingdom
- October 2025 *Advanced Data Science* (undergraduate): syllabus, materials, and projects until the course ended in 2025.
- May 2019 **Research Assistant and Research Fellow**, Trinity College Dublin, Ireland
- February 2021 Smart-city service discovery, edge service placement, and self-adaptive systems.
- March 2015 **Teaching Assistant**, Trinity College Dublin, Ireland
- February 2021 Master's and undergraduate courses in software engineering and scalable computing.
- 2011 **Earlier software and research roles**, Colombia
- 2014 Software development and graduate research (Los Andes University, Katary Software Factory).

Awards and Scholarships

Awards

November 2018 **Ph.D. Final Year Trinity Employability Award**, in partnership with Intel. Dublin, Ireland

November 2011 **Best B.Sc. Research Final Project**, Universidad de Nariño, Pasto, Colombia

Scholarships

March 2015 **Ph.D. research studentship in Dynamic Service Adaptation, Science Foundation Ireland**, Dublin, Ireland

Supervision

PhD Students

October 2025 **Harry Allen**, UKRI AI Centre for Doctoral Training in Decision Making for Complex Systems, University of Cambridge, UK, Co-supervisor: Neil Lawrence
Current Project topic: *Interpretable Self-Sustaining Systems*.

Master Students

October 2025 **Ilai Shraga**, MPhil in Advanced Computer Science, University of Cambridge, UK, Co-supervisor: Neil Lawrence
June 2026 Thesis: *SAGE-LOOP: A Reproducible Benchmarking Platform for GenAI-Augmented Self-Adaptive Systems*.

Undergrad Students

2026-Current **Emily Kelt**, University of Cambridge, UK, Co-supervisor: Neil Lawrence
Project topic: *aICU: healthcare AI systems*.

2026-Current **Preesha Dige**, University of Cambridge, UK, Co-supervisor: Neil Lawrence
Project topic: *Multi-agent systems and DOAgent*.

2024-2025 **Joseph Poon**, University of Cambridge, UK, Co-supervisor: Neil Lawrence
Project topic: *Adapting Service Placement Algorithms for Edge Computing: Implementing a Decentralized Approach in iFogSim*.

2024-2025 **Arvind Raghu**, University of Cambridge, UK, Co-supervisor: Neil Lawrence
Project topic: *CDN Simulator for Benchmarking Self-Adaptive System Approaches*.

Service to the Scientific Community

Community roles

Current **Reviewer for international peer-reviewed journals**, *IEEE Transactions on Services Computing (TSC)*, and *IEEE Internet of Things Journal*

2026 **Reviewer**, *25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*

2026 **Co-organiser**, *The Fourth UK AI Conference 2026*, UK-AI community

2026 **General Chair**, *The Sorrento Meeting 2026*, University of Cambridge

2025 **Editor**, *Proceedings of Machine Learning Research, Volume 295*, PMLR

2025 **Co-organiser**, *The 7th Workshop on Data, Learning and Inference (DALI 2025)*, DALI Sorrento 2025

2025 **Web Chair**, *20th International Conference on Software Engineering for Adaptive and Self-Managing Systems*, SEAMS 2025

2022 – 2025 **Co-organiser**, *UK AI Conference series (2023–2025)*, *NeurIPS at Cambridge Meetup (2022–2024)*, *NeurIPS Deploying ML Systems Workshop (2022)*, *ATI AI Fellows Day (2022)*

Selected Publications

Full publication list: [Google Scholar profile](#).

Peer-Reviewed Journals

- J7 **Cabrera C.**, Paleyes A., Thodoroff P., Lawrence ND., [Machine Learning Systems: A Survey from a Data-Oriented Perspective](#). *ACM Computing Surveys (CSUR)*, Vol. 58, No. 5, 2025.
- J6 Robinson D., **Cabrera C.**, Gordon AD., Lawrence ND., Mennen L., [Requirements are All You Need: The Final Frontier for End-User Software Engineering](#). *ACM Transactions on Software Engineering and Methodology (TOSEM)*, Vol. 34, No. 5, 2025.
- J5 Kazmi A., Staffolani A., Zhang T., **Cabrera C.**, Clarke S., [Dynamic Service Placement in Edge Computing: A Comparative Evaluation of Nature-Inspired Algorithms](#). *IEEE Access*, 2024.
- J4 **Cabrera C.**, Svorobej S., Palade A., Kazmi A., Clarke S., [MAACO: A Dynamic Service Placement Model for Smart Cities](#). *IEEE Transactions on Services Computing (TSC)*, 2023.
- J3 **Cabrera C.**, Clarke S., [A Self-Adaptive Service Discovery Model for Smart Cities](#). *IEEE Transactions on Services Computing (TSC)*, Vol. 15, No. 1, pp. 386–399, 2022.
- J2 Tabatabaee H., Rasool S., Kazmi A., Palade A., **Cabrera C.**, White G., Clarke S., [Dynamic Service Placement in Multi-access Edge Computing: a Systematic Literature Review](#). *IEEE Access*, Vol. 10, pp. 32639–32688, 2022.
- J1 Rojas E., Bastidas V., **Cabrera C.**, [Cities-Board: A Framework to Automate the Development of Smart Cities Dashboards](#). *IEEE Internet of Things Journal*, Vol. 7, No. 10, pp. 10128–10136, 2020.

Selected Conference Proceedings and Artifacts

- C9 Paleyes A., Sendyka R., Robinson D., **Cabrera C.**, Lawrence ND., [Prompt Variability Effects on LLM Code Generation](#). *LLM4Code 2026*, Third International Workshop on Large Language Models for Code, co-located with ICSE 2026.
- C8 Poon J., **Cabrera C.**, Lawrence ND., [iFogSim-Placement: A Simulation Framework for Edge Service Placement](#). *SEAMS 2026*, co-located with ICSE 2026 (artifact).
- C7 **Cabrera C.**, Bastidas V., Schooling J., Lawrence ND., [The Systems Engineering Approach in Times of Large Language Models](#). *HICSS*, 2025.
- C6 **Cabrera C.**, Paleyes A., Lawrence ND., [Self-sustaining Software Systems \(S4\): Towards Improved Interpretability and Adaptation](#). *SATrends* 24, 2024.
- C5 Cardozo N., Dusparic I., **Cabrera C.**, [Prevalence of Code Smells in Reinforcement Learning Projects](#). *AI Engineering (CAIN)*, 2023.
- C4 Paleyes A., **Cabrera C.**, Lawrence ND., [An Empirical Evaluation of Flow-Based Programming in the Machine Learning Deployment Context](#). *AI Engineering (CAIN)*, 2022.
- C3 Paleyes A., **Cabrera C.**, Lawrence ND., [Towards Better Data Discovery and Collection with Flow-Based Programming](#). *NeurIPS DCAI Workshop*, 2021.
- C2 **Cabrera C.**, Clarke S., [A Reinforcement Learning-Based Service Model for the Internet of Things](#). *ICSOC*, 2021.
- C1 **Cabrera C.**, White G., Palade A., Clarke S., [The Right Service at the Right Place: A Service Model for Smart Cities](#). *PerCom*, 2018.

References

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| Line manager and research supervisor | Prof. Neil D. Lawrence , <i>DeepMind Professor of Machine Learning at University of Cambridge</i> , Department of Computer Science and Technology, University of Cambridge, William Gates Building, 15 JJ Thomson Avenue, Cambridge
Email: ndl21@cam.ac.uk |
| Senior colleague | Dr. Carl Henrik Ek , <i>Associate Professor at University of Cambridge</i> , Department of Computer Science and Technology, University of Cambridge, William Gates Building, 15 JJ Thomson Avenue, Cambridge
Email: che29@cam.ac.uk |
| Ph.D. Thesis supervisor | Prof. Siobhán Clarke , <i>Professor at Trinity College Dublin</i> , School of Computer Science and Statistics, College Green, Dublin 2, Ireland
Email: Siobhan.Clarke@scss.tcd.ie |